

Syed Azlan zar

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WORK EXPERIENCE

Zenkoders

Full stack Developer

Karachi, Pakistan

Feb 2024 – Apr 2025

- Built and maintained backend services (Node.js, NestJS, MongoDB), achieving 99.9% uptime and reducing API response times by 40% for 5+ products.
- Contributed to optimizing backend logic and improving query performance, increasing data efficiency by 35% and supporting large-scale user loads.
- Contributed to 10+ backend projects, adding new features and fixing critical bugs, reducing issue resolution time by 50%.
- Integrated OpenAI APIs with advanced prompt engineering, improving content accuracy by 25% and enhancing user satisfaction for 1,000+ active users.
- Implemented CI/CD pipeline (GitHub Actions, Vercel), accelerating release cycles by 30% and enabling weekly deployments.
- Conducted 100+ API tests using Postman, ensuring 98% test coverage and minimizing integration errors by 40%.
- Collaborated with frontend teams on 15+ feature rollouts, improving cross-team efficiency and reducing handover delays by 20%.

Zenkoders

Frontend Developer

Karachi, Pakistan

Oct 2022 – Jan 2024

- Built 70+ high-performance UIs with JavaScript, React, Next.js, TypeScript, and Tailwind CSS, increasing user satisfaction by 30% and supporting 50K+ monthly visitors.
- Translated complex designs into responsive, accessible frontends, reducing design-to-release time by 25% and improving accessibility scores by 20%.
- Integrated REST APIs in 20+ projects, improving load speed by 30% and reducing bounce rates by 20%.
- Ensured frontend reliability with 200+ test cases in Postman and dev tools, reducing frontend errors by 35%.
- Enhanced code quality and reduced bugs by 40% through strict TypeScript practices across all projects.
- Achieved 85–90% Lighthouse scores on key pages, increasing organic traffic by 25% and boosting SEO ranking.

SKILLS & INTERESTS

Languages: JavaScript, TypeScript, Python, Node.js and C++ (Intermediate/Academic).

Frameworks: Next.js, React.js, NestJS, FastAPI, Strapi, Redux, React Query, Numpy, Pandas, PyTorch.

AI & Data Science: Retrieval-Augmented Generation (RAG), Microsoft Presidio (NLP Sanitization), ChromaDB (Vector Store), LLMs (Google Gemini/OpenAI APIs), Neural Networks, Transformer Models.

Databases: PostgreSQL, MongoDB, ChromaDB, SQL and Firebase Firestore.

Tools: Git, GitHub Actions, Vercel, Docker (basic), Postman, Stripe, Jira, Slack.

Architecture: Multi-Tenant Architecture, Microservices, REST APIs, Object-Oriented Programming (OOP).

EDUCATION AND TRAINING

Brandenburg University of Technology

Masters in Artificial Intelligence

Cottbus, Germany

2025 - 2027

- Goal is to learn the core foundation of AI, Neural Networks and Data Cycle related Data Engineering, Big Data.
- Analytics, LLM, Transformer Model, RNN, LSTM, RL, Agentic AI.
- Machine Learning frameworks like Torch/TensorFlow for Autonomous Systems.

University of Karachi

Bachelor's of Science in Computer Science

Karachi, Pakistan

2019 - 2023

- Snappy Kitchen is a subscription-based meal delivery platform that lets users manage weekly or monthly meal plans, while vendors and admins handle menus, orders, and subscriptions through dedicated portals. Built with Next.js, Vue.js, NestJS, Prisma, PostgreSQL, RabbitMQ, and a microservices architecture for scalable and efficient operations.

PROJECTS

PrivRAG | Privacy-Preserving Multi-Tenant RAG System

- **Architecture:** Engineered a secure, multi-tenant platform featuring a Next.js frontend, a NestJS main API gateway, and a high-performance Python/FastAPI AI microservice.
- **Data Isolation & Security:** Architected strict logical data isolation within a persistent ChromaDB vector store, implementing isolated collections per tenant to ensure total cross-organizational data privacy.
- **NLP Data Sanitization:** Developed a deterministic and heuristic hybrid NLP masking pipeline using Microsoft Presidio and custom regular expressions to anonymize complex PII (Names, IDs, custom VPN keys) before vector embedding or LLM parsing.
- **Orchestration & Sync:** Streamlined real-time document chunking, metadata tracking, and secure Gemini LLM API reasoning, utilizing a custom-built data-wiping routine to synchronize relational state deletions with the vector database.

AI Cover Letter Generator | Chrome Extension & Microservices Architecture

- **Frontend Extension:** Developed an AI-powered Chrome Extension using React, TypeScript, Vite, and Tailwind CSS, allowing users to securely authenticate, upload PDF resumes, and input job preferences directly from their browser.
- **Backend Orchestration:** Engineered a NestJS REST API gateway with MongoDB (Mongoose) and strict JWT-based authentication to manage user sessions, coordinate secure media uploads, and render server-side EJS templates.
- **AI & NLP Parsing:** Built a FastAPI Python microservice to extract structured JSON data from complex PDF resumes and utilized Hugging Face Transformer models to generate highly personalized, context-aware cover letters.
- **End-to-End System Design:** Architected a distributed, 3-tier system seamlessly handling cross-origin communication between the browser extension, the Node.js backend, and the Python AI inference engine, culminating in automated PDF generation and delivery.

Vector Space Clustering & Classification | Core Machine Learning Algorithms

- **Algorithmic Engineering:** Developed mathematical, from-scratch Python implementations of fundamental machine learning algorithms (K-Means, DBSCAN, and K-Nearest Neighbors) to analyze spatial data without relying on high-level libraries.
- **Computer Vision Segmentation:** Engineered an automated image segmentation pipeline that maps image pixels into 3D RGB vector spaces, utilizing custom clustering models to group color centroids and output segmented visual data.
- **Supervised Learning Benchmarks:** Designed and executed a classification benchmarking suite to evaluate the custom-built KNN model against industry-standard frameworks using complex scientific datasets (Breast Cancer and Iris).
- **Multi-Dimensional CLI Architecture:** Built a scalable Command Line Interface (CLI) tool equipped to process n-dimensional synthetic datasets (2D, 3D, 5D), proving the mathematical universality of the clustering logic across high-dimensional vector spaces.

HOBBIES & CREATIVITY

- *Photography*
- *Digital Content Creation*